

Getting Started with Oracle VM VirtualBox

Here's an article that introduces you to the joys of virtualisation by taking you through the fairly simple steps of setting up Oracle's virtual machine—VirtualBox.



We often feel the need to access multiple operating systems. Though you can install multiple operating systems on the same hard disk at different locations, you can access only one OS at a time, resulting in a lot of computing resources being wasted. Virtualisation is the best solution for this problem, in which a virtual machine is installed on the host OS. On this virtual machine, various operating systems can not only be installed as guests, but also can be accessed at the same time. Oracle's VM, VirtualBox, is one such cross-platform virtualisation application. It can be easily installed on any existing Intel or AMD-based computer running Windows, Mac, Linux or Solaris (called the host OS), and can run multiple operating systems (inside multiple virtual machines) that are called guest OSs, all at the same time.

An introduction

VirtualBox runs on a large number of 32-bit and 64-bit host operating systems. It can present up to 32 virtual CPUs to each virtual machine, irrespective of the number of CPU cores present physically on the host system. It also implements a virtual USB controller and allows you to connect arbitrary USB devices to your virtual machines without having to install device-specific drivers on the host.

You will find that VirtualBox can adjust many times higher resolution than that of a physical screen, allowing VM to be spread over a large number of screens attached to the host system. It has a feature that can save arbitrary snapshots of the state of the virtual machine. You can go back in time and revert the virtual machine to any such snapshot and start an alternative VM configuration from there, effectively creating a whole snapshot tree. In this article we will be going through the steps for creating a virtual machine in VirtualBox 4.1.8 for Ubuntu 11.10, configuring Fedora12 as the guest OS.

Getting started

Oracle provides the link <http://www.virtualbox.org> from where you can download VirtualBox 4.1.8 for different OSs. Download the set-up for your OS. As we are considering the Ubuntu 11.10 as the host OS, go to VirtualBox 4.1.8 for Linux hosts, and download VirtualBox for Ubuntu 11.10. After you have installed it, create your first virtual machine by clicking the **New** button on the left side corner as indicated in Figure 1. The *New Virtual Machine* wizard gets started; go through it and select the operating system you want to install in it from the given options. If the OS you want to install is not among the options offered, you can opt for other options. But as we now want to configure Fedora 12 for our virtual machine, select corresponding options and give a self-descriptive name for your virtual machine—something like *Fedora-Linux*, which will be displayed on the home page of VirtualBox.

Next, it is recommended that the RAM allocated to this virtual machine should be set according to the RAM of your physical system. Suppose your RAM is 2 GB and you are going to create one or more virtual machine(s) inside VirtualBox, then it is best that you allocate 512 MB to each virtual machine. This is because when your virtual machine is started, the part of RAM allocated to that guest OS becomes unavailable from your host OS. If the number of virtual machines increases, RAM from the host system decreases further, thereby decreasing the performance of your system and in the worst cases, causes your host OS to hang. So RAM should be allocated according to your needs and the resources available.

Next, you have to allocate a virtual hard disk to your virtual machine. Here, you have two options—to either create a new hard disk or use the existing one with a minimum of 8 GB of space. But if this is your first virtual machine, then you need to create a new one. (Refer to